Compulsory Exercise

Topics in Life Insurance (Liv2), Block 3, 2026

March 18, 2026

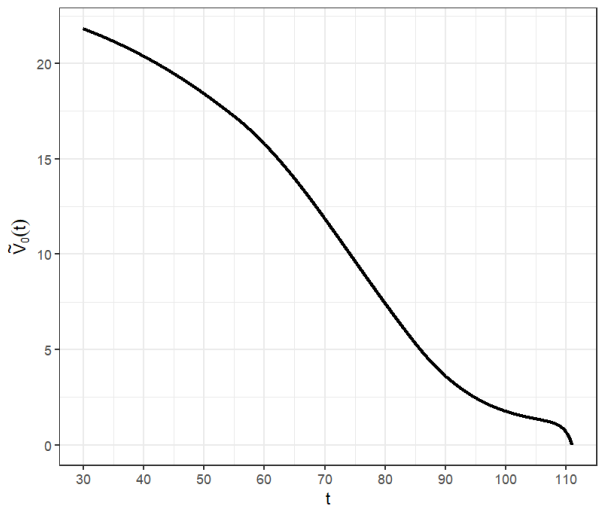
(i)

Using Lebniz' rule we obtain af ordinary backward differential equation for the reserve $\tilde{V}_0$. We have

$$\begin{aligned}
\frac{d}{dt}\tilde{V}_0(t) &= \int_t^\infty \frac{\partial}{\partial t} e^{-\int_t^s (\tilde{r}(u)+\tilde{\mu}(u))\,du}\,ds - 1 \\
&= \int_t^\infty e^{-\int_t^s (\tilde{r}(w)+\tilde{\mu}(w))\,dw}(\tilde{r}(t)+\tilde{\mu}(t))\,ds - 1 \\
&= (\tilde{r}(t)+\tilde{\mu}(t)) \int_t^\infty e^{-\int_t^s (\tilde{r}(w)+\tilde{\mu}(w))\,dw}\,ds - 1 \\
&= (\tilde{r}(t)+\tilde{\mu}(t))\tilde{V}_0(t) - 1,
\end{aligned}$$

with $\tilde{V}_0(111) = 0$.

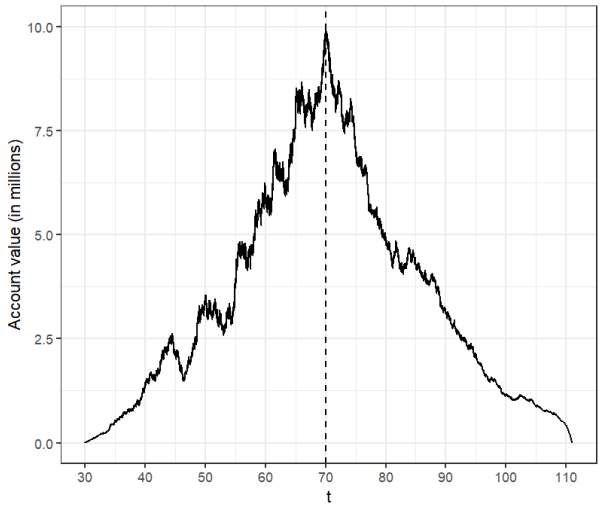
Now, using the fourth order Runge-Kutta method we optain a numerical solution of $\tilde{V}_0(t)$. The result is shown via the plot below



Exercise (ii)

(iii)

A simulation of the account is provided below.



We observe from our simulation that it begins from age $t = 30$ at the value of zero as demanded. Thereafter, the account accumulates until retirement $m = 70$ with a value of around 9.8 million (monetary units). Then the account de-cumulates until our assumed lifetime cut-off at $t = 111$. We also observe random fluctuates due to the fact that we model the risky asset via a Brownian motion.

Exercise (iv)

We argue that $V^0(t,x) = x$ by verification: we show that $V^0(t,x) = x$ satisfies Thiele's partial differential equation (PDE) in both phases with the correct terminal boundary condition, and conclude by uniqueness.

Thiele's PDE

The market reserve $V^0(t,x)$ is defined as the expected discounted future net outflows,

$$V^0(t,x) = \mathbb{E}^{\mathbb{P}\otimes\mathbb{Q}}\left[\int_t^{111} e^{-\int_t^s (\tilde{r}(v)+\mu(v))\,dv}\, c(s,X(s))\,ds \;\middle|\; X(t)=x, Z(t)=0\right].$$

By the Feynman–Kac theorem—equivalently, by applying Itô's formula to the discounted process $e^{-\int_0^t (\tilde{r}+\mu)\,ds} V^0(t,X(t))$ and requiring its drift to equal the outflow rate—$V^0(t,x)$ satisfies Thiele's PDE

$$\frac{\partial V^0}{\partial t} + a(t,x)\,\frac{\partial V^0}{\partial x} + \tfrac{1}{2}\,b(t,x)^2\,\frac{\partial^2 V^0}{\partial x^2} = \big(\tilde{r}(t)+\mu(t)\big)V^0 - c(t,x), \tag{1}$$

where $a(t,x)$ and $b(t,x)$ are the drift and diffusion of X, and $c(t,x)$ is the net outflow rate (benefits minus premiums).

Derivatives of $V^0(t,x) = x$

We conjecture $V^0(t,x) = x$ and compute its partial derivatives:

$$\frac{\partial V^0}{\partial t} = 0, \qquad \frac{\partial V^0}{\partial x} = 1, \qquad \frac{\partial^2 V^0}{\partial x^2} = 0.$$

Substituting into (1), the PDE reduces to verifying

$$a(t,x) = \big(\tilde{r}(t)+\mu(t)\big)x - c(t,x). \tag{2}$$

Retirement phase, $t \in [m, 111]$

From SDE (2), the drift coefficient is

$$a(t,x) = \big(\tilde{r}(t)+\mu(t)\big)x - \frac{x}{\tilde{V}^0(t)},$$

There is no death benefit during retirement (the account is released to the pool on death), so the net outflow rate consists solely of the annuity payout, $c(t,x) = x/\tilde{V}^0(t)$.

We need to verify (2), i.e. that $a(t,x) = \big(\tilde{r}+\mu\big)x - c(t,x)$. Equivalently, we show that the left-hand side minus the right-hand side is zero:

$$\begin{aligned}
&a(t,x) - \big(\tilde{r}+\mu\big)x + c(t,x) \\
&= \left[\big(\tilde{r}+\mu\big)x - \frac{x}{\tilde{V}^0(t)}\right] - \big(\tilde{r}+\mu\big)x + \frac{x}{\tilde{V}^0(t)} \\
&= 0. \quad \checkmark
\end{aligned}$$

Active phase, $t \in [30, m)$

From SDE (1), the drift is

$$a(t,x) = \tilde{r}(t)\,x + \theta L(t) - \mu(t)\big(KL(t) - x\big)$$

During the active phase the policyholder pays premiums at rate $\theta L(t)$ (an inflow to the insurer) and the insurer pays death benefits at expected rate $\mu(t)\,KL(t)$. The net outflow rate is therefore

$$c(t,x) = \mu(t)\,KL(t) - \theta L(t).$$

We verify (2) in the same way as above:

$$\begin{aligned}
&a(t,x) - \big(\tilde{r} + \mu\big)x + c(t,x) \\
&= \big[(\tilde{r} + \mu)x + \theta L - \mu KL\big] - (\tilde{r} + \mu)x + \big[\mu KL - \theta L\big] \\
&= 0. \quad \checkmark
\end{aligned}$$

Boundary condition

At the terminal age $t = 111$ there are no future obligations, so the boundary condition is $V^0(111, x) = 0$. If $V^0(t,x) = x$, this requires $X(111) = 0$. This is guaranteed by construction of the calculation basis: the reserve $\tilde{V}^0(t)$ computed in Exercise (i) satisfies $\tilde{V}^0(111) = 0$, so the annuity payout rate $X(t)/\tilde{V}^0(t) \to \infty$ as $t \to 111$, which drains the account completely. In other words, the purpose of $\tilde{V}^0$ is precisely to calibrate the payout rate so that the account is exhausted at the maximum age. The simulation from Exercise (ii) confirms this numerically: $X(111) \approx 0$.

Conclusion

Since $V^0(t,x) = x$ satisfies Thiele's PDE (1) in both phases and matches the terminal boundary condition, uniqueness of the solution gives

$$\boxed{V^0(t,x) = x.}$$

The market reserve equals the account value. This is the defining property of a unit-link product: the policyholder bears all investment risk, and the account dynamics are constructed to continuously budget for every mortality and benefit cash flow, making the account the exact self-financing replicating portfolio for all future obligations.

Exercise (v)

For $t \geq m$, we can see from the account dynamics, $dX(t)$, that the life annuity benefit process is

$$Y(t) = b^0(t, X(t)) = \frac{X(t)}{\tilde{V}_0(t)}.$$

Since $\tilde{V}_0(t)$ is deterministic, we apply the product rule to $Y = X \cdot (1/\tilde{V}_0)$.

Ingredient 1: SDE for X

From SDE (2), for $t \geq m$:

$$\mathrm{d}X(t) = \left[(\pi(t)\alpha + (1-\pi(t))r + \mu(t))X(t) - \tfrac{X(t)}{\tilde{V}_0(t)}\right] \mathrm{d}t + \sigma\pi(t)X(t)\,\mathrm{d}W(t).$$

Ingredient 2: ODE for $\tilde{V}_0$

From the Thiele ODE derived in Exercise (i):

$$\tilde{V}_0'(t) = \big(\tilde{r}(t) + \tilde{\mu}(t)\big)\tilde{V}_0(t) - 1,$$

so that

$$\mathrm{d}\left(\frac{1}{\tilde{V}_0(t)}\right) = -\frac{\tilde{V}_0'(t)}{\tilde{V}_0(t)^2}\,\mathrm{d}t = \left[-\frac{\tilde{r}(t) + \tilde{\mu}(t)}{\tilde{V}_0(t)} + \frac{1}{\tilde{V}_0(t)^2}\right]\mathrm{d}t.$$

Product rule

Applying the product rule $\mathrm{d}Y = \frac{1}{\tilde{V}_0}\,\mathrm{d}X + X\,\mathrm{d}\big(\frac{1}{\tilde{V}_0}\big)$ (no cross-variation term since $\tilde{V}_0$ is deterministic), we get:

$$\frac{1}{\tilde{V}_0(t)}\,\mathrm{d}X(t) = \Big[(\pi(t)\alpha + (1-\pi(t))r + \mu(t))\,Y(t) - \frac{X(t)}{\tilde{V}_0(t)^2}\Big]\mathrm{d}t + \sigma\pi(t)\,Y(t)\,\mathrm{d}W(t),$$

$$X(t)\,\mathrm{d}\Big(\frac{1}{\tilde{V}_0(t)}\Big) = \Big[-(\tilde{r}(t) + \tilde{\mu}(t))\,Y(t) + \frac{X(t)}{\tilde{V}_0(t)^2}\Big]\mathrm{d}t.$$

The $X(t)/\tilde{V}_0(t)^2$ terms cancel, giving

$$\begin{aligned}
\mathrm{d}Y(t) &= \big(\pi(t)\alpha + (1-\pi(t))r - \tilde{r}(t) + \mu(t) - \tilde{\mu}(t)\big)\,Y(t)\,\mathrm{d}t + \sigma\pi(t)\,Y(t)\,\mathrm{d}W(t),\\
Y(m) &= \frac{X(m)}{\tilde{V}_0(m)}.
\end{aligned}$$

This is a (generalised) geometric Brownian motion since both the drift and diffusion coefficients are of the form $f(t)Y(t)$.

Interpretation of the calculation basis

The calculation basis $(\tilde{r}(t), \tilde{\mu}(t))$ determines the drift of $Y(t)$ and thereby the expected payout profile. With the specific choice from Table 2, $\tilde{r}(t) = \pi(t)\alpha + (1-\pi(t))r$ and $\tilde{\mu}(t) = \mu(t)$, the drift vanishes identically:

$$\pi(t)\alpha + (1-\pi(t))r - \tilde{r}(t) + \mu(t) - \tilde{\mu}(t) = 0.$$

In this case $Y(t)$ is a driftless geometric Brownian motion—an exponential martingale—so the expected payout is constant: $\mathbb{E}[Y(t) \mid Y(m)] = Y(m)$ for all $t \geq m$. This produces a *flat* expected payout profile.

More generally:

- A more conservative (lower) $\tilde{r}(t)$ increases the drift of $Y(t)$, giving an *increasing* expected payout profile (start low, grow over time).
- A more aggressive (higher) $\tilde{r}(t)$ decreases the drift, producing a *decreasing* profile (start high, decline).
- Analogously, choosing $\tilde{\mu}(t) < \mu(t)$ adds positive drift (the basis "under-predicts" mortality), leading to increasing payouts.

The calculation basis thus acts as a lever for shaping the expected trajectory of retirement income.

Exercise (vi)

We simulate $N = 1000$ paths of the life annuity benefit $Y(t)$ for $t \in [m, 111]$ using an Euler scheme with step length $h = 1/100$ on the dynamics derived in Exercise (v):

$$\mathrm{d}Y(t) = \big(\pi(t)\alpha + (1-\pi(t))r - \tilde{r}(t) + \mu(t) - \tilde{\mu}(t)\big)\, Y(t)\,\mathrm{d}t + \sigma\pi(t)\, Y(t)\,\mathrm{d}W(t).$$

With the calculation basis from Table 2 the drift coefficient vanishes, so the Euler step reduces to

$$Y(t_{i+1}) = Y(t_i) + \sigma\,\pi(t_i)\,Y(t_i)\,\sqrt{h}\,Z_i, \qquad Z_i \overset{\text{iid}}{\sim} N(0,1).$$

The initial value is $Y(m) = X(m)/\tilde{V}_0(m)$, where $X(m) \approx 9{,}821{,}181$ is taken from the simulated path in Exercise (ii), giving $Y(70) \approx 827{,}421$. The dashed line in the figure marks this initial benefit level. A selection of 15 sample paths is shown below.

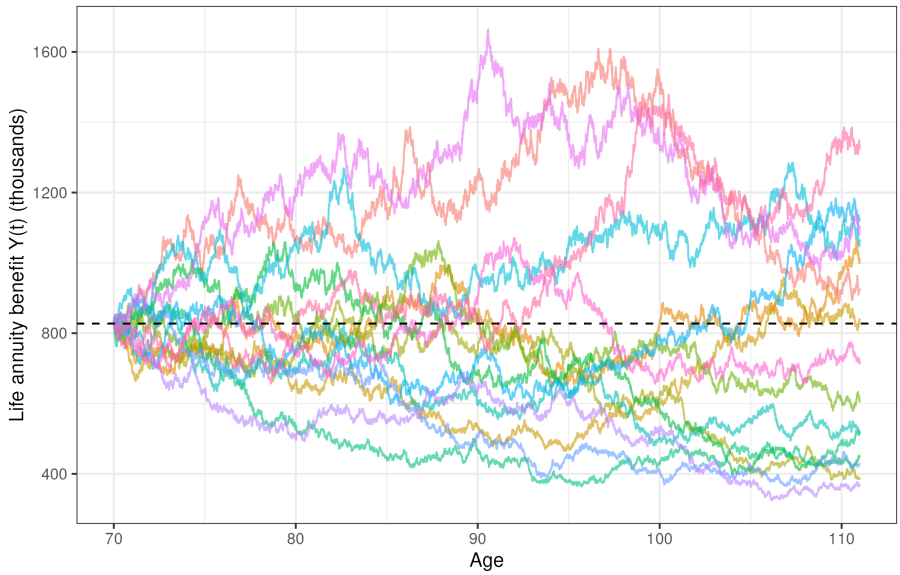


Figure 1: 15 simulated paths of the life annuity benefit $Y(t)$. The dashed line marks the initial benefit $Y(m)$.

We observe significant variation across paths: some paths see benefits double or more, while others decline substantially. This dispersion reflects the stochastic investment risk borne by the insured under the unit-link design.

Exercise (vii)

Based on the $N = 1000$ simulated paths from Exercise (vi), we estimate the expected life annuity benefit $\mathbb{E}[Y(t) \mid X(m)]$ at each time point by the sample mean across paths:

$$\hat{\mathbb{E}}[Y(t) \mid X(m)] = \frac{1}{N} \sum_{n=1}^{N} Y^{(n)}(t).$$

Figure 2: Estimated expected life annuity benefit $\mathbb{E}[Y(t) \mid X(m)]$ over time. The dashed red line marks $Y(m)$.

We observe that the expected payout profile is approximately *flat* over the entire retirement period. This is consistent with the theoretical result from Exercise (v): since $\tilde{r}(t) = \pi(t)\alpha + (1 - \pi(t))r$ and $\tilde{\mu}(t) = \mu(t)$, the drift of Y is zero and Y is an exponential martingale, so $\mathbb{E}[Y(t) \mid X(m)] = Y(m)$ for all $t \geq m$. The small fluctuations visible in the plot are Monte Carlo noise from the finite sample of $N = 1000$ paths.

In practice, this flat expected profile means that, on average, the insured receives the same benefit throughout retirement. However, as seen in Exercise (vi), individual paths exhibit substantial variability around this expectation—the insured bears significant investment risk. A more conservative calculation basis (with $\tilde{r}(t) < \pi(t)\alpha + (1 - \pi(t))r$) would instead produce an increasing expected payout profile, starting lower but growing over time, which could be preferable from a product design perspective.